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TP Aeolian Dynamics Wind Farm, 10.8 MW, at Agia Anna Design, Manufacturing, Supply, Supervision of Erection and Commissioning of the MV Switchgear and its associated Control Equipment, Telecommunication Equipment, Load Flow & Short Circuit Study and Earthing Study of the 10.8MW Wind Farm in Ag. Anna, Larnaka, Cyprus. Project: TP Aeolian Dynamics Wind Farm, 10.8MW at Ayia Anna				
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1. General

This document provides the necessary data concerning the relay settings for the IEDs triggering the circuit breakers in the MV Boards, of the TP Aeolian Dynamics Cyprus 10.8MW wind farm (TP Aeolian Dynamics Ltd).

2. Reference Documents

1. Project specifications – “EAC Specification 14-019”, Issue 05, dated 10/06/2014.
2. Project specifications – “T.P. AEOLIAN DYNAMICS WIND PARK FOR T.P. AEOLIAN DYNAMICS LTD AT AYIA ANNA/ Electrical Installations Volume B/ Electrical Specification”, ELIOFOTOU, ZINIERIS & STASIS PARTNERS.
3. IEC 60909 – (0 to 4) – “Short-circuit currents in three-phase a.c. systems”.
4. IEC 60255 – 3 – “Single input energizing quantity measuring relays with dependent or independent time”.
5. IEC 60364 – 4 - 43 – “Electrical installations of buildings – Protection for safety – Protection against overcurrent”.
6. IEC 60947 – (1 to 3) – “Low voltage switchgear and controlgear”.
7. Document – “AYIA ANNA WIND FARM LOAD FLOW.pdf”, e-mail from 05/12/2014, Papachristoforou Christakis.
8. Document – “AS-site-plan-new Model (1).pdf”, e-mail from 06/12/2014, Papachristoforou Christakis.
9. Document – “Material Submissions Ayia Anna e01.pdf”, e-mail from 06/12/2014, Papachristoforou Christakis.
10. Document – “MV CABLES LENGTH.pdf”, e-mail from 06/12/2014, Papachristoforou Christakis.
11. Document – “cable.xls”, e-mail from 05/12/2014, Papachristoforou Christakis.
12. Document – “SERVER-468RG29V9258.pdf”, e-mail from 05/12/2014, Papachristoforou C.
13. Document – “SERVER-468RG29V11942.pdf”, e-mail from 05/12/2014, Papachristoforou C.
14. Document – “IG-159-GB-06.pdf”, e-mail from 16/12/2014, Kairis Stamatios.
15. Document – “V100-2MW_SWITCHGEAR-SETTINGS_R00.pdf”, e-mail from 16/12/2014, Kairis Stamatios.
16. Document – “REQUIRED DATA Agia Anna Wind Farm for T P Aeolian Dynamics Ltd_SP.XLS”, e-mail from 16/12/2014, Kairis Stamatios.
17. Data for 1,8MW (Vestas V-100) – Internal Transformer & Cable between Motor and Transformer & Transformer and Wind Generator MV Board, e-mail from 19/12/2014, Kairis Stamatios.
18. Data for Overhead Line between Psevdas S/S and TP Aeolian S/S – “line aeolian.xls”, e-mail from 22/12/2014, Papadouris Savvas.
19. Data for Minimum & Maximum Short Circuit Power and Current, e-mail from 22/12/2014, Papadouris Savvas.

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20. Document – “Zinieris 9_9_14 single lines drawings for the control and protection of the MW SG.pdf”, 09/09/2014, Zinieris Kyriakos.
21. Drawing – “E-01.pdf”, Drawing No: E-01, 06/10, Rev A.
22. Drawing – “E-02.pdf”, Drawing No: E-02, 06/10, Rev -.
23. Documents – “=AA1J00+J01.pdf, =AA1J00+J02.pdf, =AA1J00+J03.pdf, 1VB2383201_001_00.pdf, 1VB2383202_001_00.pdf & 1VB2383203_001_00.pdf”, e-mails from 22/10/2014 & 24/10/2014, ABB Calor Emag.
24. Short Circuit Study software file – “3VGR111416E0002_-.nepprj” – REV-, 08/01/2015.
25. Document – “630 Series Technical Manual”, Rev D, Doc. No: 1MRS756508, 18/09/2012.

3. Data - Assumptions

The calculations have been performed with the aid of NEPLAN Version 5.3.4 software, which conforms to the IEC calculation standards. The short circuit calculation study has been performed according to the IEC 60909 standard and the load flow analysis according to the extended Newton Raphson method.

3.1 Data

- 1) The technical data for network feeder were elicited from documents [19] & [22] of A. Eracleous Electrical Installations Ltd calculations. The data used are shown at Annex A.
- 2) The technical data for 22kV MV cables were elicited from documents [14] & [16]. Due to the fact that zero sequence capacitance was not given for documents [14] & [16] it was assumed that zero sequence capacitance is equal to positive sequence capacitance. The data used are shown at Annex A.
- 3) The technical data for the wind generator transformers were given from the manufacturers in document [17] (quoting “we confirm that the data are the same as in Koshi WF”) and information was also extracted from documents [7] & [22]. The data used are shown at Annex A.
- 4) The technical data for the 1.8MW asynchronous wind generators were drawn out of document [16]. The data used are shown in Annex A.
- 5) The REF 630 IED incorporates Lockout (86 – ANSI) and TCS – Trip Coil Supervision functions. However the aforementioned protection functions are not described in this document as irrelevant to the selectivity matters covered.

Further data concerning the protective devices used in the MV Switchgear panel of each wind generator were taken from documents [16], [15] & [14]. Next data concerning the MV Switchgear panel (circuit breakers, current transformers and panel busbar electrical data) at the TP Aeolian Dynamics S/S were taken from documents [23]. Furthermore other useful data were extracted from the load flow study [7], the recommended relay protection settings from Vestas [15] and the single line diagram [22]. Finally technical data concerning the substation topology, cable lengths and cable routing were drawn out from document [21].

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3.2 Assumptions

- 1) 1) Due to lack/ mismatch of technical data for the M.V. cable 3x1x300mm² Al 22kV, the electrical data given from document [18] were used and modified according to reference document [3]. The abovementioned cable was assumed to be buried in the ground and its electrical data are presented in Annex A.
- 2) 2) Due to mismatch of technical data for M.V. cables EPR or HEPR 3x70/70mm² Cu 22kV, the values of the zero sequence resistance and inductance were calculated according to document [3] and are presented in Annex A. The aforementioned cables are assumed to be routed in air.
- 3) Due to inconsistency between the given technical data from document [17] and the values calculated automatically by NEPLAN software, as well as by hand, the simulation was made with the data presented in Annex A.
- 4) The maximum short circuit calculations were made according to standard IEC 60909 – 0/(2001) assuming that all switches and breakers were closed (no open points in the examined network). Furthermore it was assumed that all nodes have 110% of their nominal voltage at the short circuit start and the network feeder and asynchronous machines contribute to the short circuit with their maximum capacity.
- 5) The minimum short circuit calculations were made according to standard IEC 60909 – 0/(2001) assuming that all switches and breakers were closed (no open points in the examined network). Furthermore it was assumed that all nodes have 90% of their nominal voltage at the short circuit start and the network feeder contributes to the short circuit. Asynchronous machines contribution to the short circuit is neglected by NEPLAN, according to IEC standard guidelines. Additionally it has to be mentioned that the resistance of the lines (cables) are taken by increased temperature.
- 6) Due to lack of further technical data for the starting time of the wind generators, it is assumed that the starting time is approximately 5 sec. The aforementioned data is elicited from document [3], for reversible static converter – fed drives feeding big fans.
- 7) Due to the lack of technical data two load flow scenarios were examined. For the incomer board relay setting it was assumed that five out of six wind generators are operating at full load and one is starting. For the feeder boards it was assumed that two wind generators are working at full load and one is starting.
- 8) The protection of the MV (22kV) network up to the MV incomer board (fed from the EAC overhead line) from short circuit current contributions, stemming from wind farm loop 1, is achieved by setting the trip threshold levels of overcurrent and earthfault functions very low. The same applies for the protection settings of wind farm feeder 2. Additionally the incorporation of directional overcurrent and earthfault protections in both wind farm feeders ensures the protection of the (22kV) MV network up the incomer board and offers information from which wind farm cluster the fault stems.
- 9) The short circuit factor of ($I_{sc} >> I_n$) of the instantaneous phase protection function of the ekorRPG202A IED, placed in each of the wind generator MV boards, will be modified to 7 times the rated current, in agreement with VESTAS. In case unwanted tripping of the IEDs is observed during the setting into function/ electrification of the MV transformer of each wind generator, the said setting will be modified so as to resolve any problem.

The technical data for cables used are shown at Annex A.

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4. Results – Relay Settings

The current versus time curves of the relay protection modules are shown in Annex C and the operation thresholds are shown in the tables at Annexes B1 & B2. According to the following relay settings, selective operation is provided to the triggered circuit breakers.

Non – Directional low-set/ high-set and instantaneous current elements for (OC) overcurrent and (EF) earthfault purposes have been activated in IEDs of the MV incomer (=AA2J00+J03) and outgoing feeders (=AA1J00+J01 & =AA1J00+J02). The purpose of the aforementioned elements in the outgoing feeders is to clear faults occurring towards and from the wind park direction.

Next directional high-set elements for (OC) overcurrent and (EF) earthfault purposes have been activated in the IEDs of the outgoing feeders at TP Aeolian Dynamics S/S. The purpose of the aforementioned elements is to clear faults occurring towards the wind park direction (contribution of each wind farm loop to the short circuit). The “START” signal of the directional (OC) “DPHHPDOC” and (EF) “DEFHPDEF” elements of each outgoing feeder (=AA1J00+J01 & =AA1J00+J02) will be blocking the non-directional elements “PHHPTOC” and “EFIPTOC” respectively, of the incomer board (=AA1J00+J03), thus achieving the reverse interlocking between outgoing feeders and incomer at TP Aeolian Dynamics substation.

The “START” signal of the non-directional (OC) “PHHPTOC” and (EF) “EFIPTOC” elements of the MV incomer board IED (=AA1J00+J03) at TP Aeolian Dynamics S/S, will be blocking the non-directional elements “PHHPTOC” and “EFIPTOC” respectively, of each outgoing feeder board (=AA1J00+J01 & =AA1J00+J02) at TP Aeolian Dynamics substation, thus achieving reverse interlocking between outgoing feeders and incomer boards.

Additionally the sensitive earth fault (51NS – ANSI) protection, with adjustable time delay is catered by the “EFLPTOC” IED function used in REF630 incomer relay. The “EFLPTOC” function offers the option of time delay if needed, while at the same time can achieve low threshold setting values of 4 and 5(A).

As presented in Annex B1, the short circuit thresholds anticipated from each loop of the TP Aeolian Dynamics wind power plant are significantly lower than the combined contribution in a short circuit event, by the EAC network and the wind farm loops.

Furthermore the circuit breaker failure protection module “CCBRBRF”/ (50BF ANSI coding) needs to be activated in REF630 relays of the incomer and feeder boards. The output “CCBRBRF operated” of the breaker failure protection module will be activated by the following protection functions of the feeder (=AA1J00+J01 & =AA1J00+J02) REF 630 relays: “PHLPTOC”, “PHHPTOC”, “DPHHPDOC”, “EFHPTOC”, “EFIPTOC” and “DEFHPDOC”. In case of breaker failure in the outgoing feeder boards (=AA1J00+J01 & =AA1J00+J02), the “CCBRBRF” protection module will trigger the circuit breaker of the incomer board (=AA1J00+J03).

It should also be noted that according to the selected phase full load current value (61 (A)) the phase and earth protection functions of the ecorRPG-202A relay offer selectivity towards the protection offered by the feeder and incomer relays. Furthermore for 61(A) selected as base current value then the cable interconnecting the asynchronous wind generator machine and the wind generator step-up transformer is protected adequately as the cable has a continuous current ability of 185,6 (A), and the step-up transformer a nominal operating current of 61 (A). It is thus suggested that the parameters of the ecorRPG-202A relay are entered manually and as installation transformer power P_t , the value of 2.1(MVA) is used, which is the maximum wind generator active power and allows the protection of the MV windings of the step-up wind generator transformer.

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Additionally the protection settings of the relay placed in the MV feeder panel of TP Aeolian Dynamics (=J04+J04), situated at Psevdas substation MV room are presented in the form of suggested settings in this study.

The settings are illustrated below in more detail.

IED Type: REF630(TP AEOLIAN DYNAMICS WINDFARM FEEDER AT PSEVDAS)

This IED provides overcurrent & earth fault protection for the MV network at Psevdas S/S and the overhead line connecting TP Aeolian Dynamics wind farm to Psevdas S/S. The aforementioned IED also provides AC reclosing, distance and breaker failure functions. However only the proposed overcurrent and earthfault protection functions are presented in this document. The basic IED function settings that are proposed for the said application are illustrated in the latter tables:

Overload Protection (51P-1 ANSI) - PHLPTOC

	Unit	Step	Value
Start Value:	i/pu	0,01	0,05-5,0
Time Multiplier (range):	t/pu	0,05	0,05-15,00
Operation:	Off / On	-	On
Num of start phases:	1 out of 3 2 out of 3 3 out of 3	-	1 out of 3
Start Value Mult (...*In):	i/pu		0,77
Time Multiplier:	t/pu		0,95
Operating Curve Type:			IDMT (IEC normal inverse)

Overcurrent Protection (51P-2 ANSI) - PHHPTOC

	Unit	Step	Value
Start Value:	i/pu	0,01	0,10-40,00
Operate Delay Time:	s	0,01	0,02-200,00
Operation:	Off / On	-	On
Num of start phases:	1 out of 3 2 out of 3 3 out of 3	-	1 out of 3
Start Value Mult (...*In):	i/pu		1,83
Time Delay:	s		0,50
Operating Curve Type:			DT (IEC Def. time)

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Earthfault Protection (51N-2 ANSI) - EFHPTOC

	Unit	Step	Value
Start Value:	i/pu	0,01	0,10-40,00
Time Multiplier (range):	t/pu	0,05	0,05-15,00
Operation:	Off / On	-	On
Start Value Mult (...*In):	i/pu		0,72
Time Multiplier:	t/pu		0,3
Operating Curve Type:			IDMT (IEC normal inverse)
Operate Delay Time:	s	0,01	0,02-200,00
Time Delay:	s		Applicable only for DT curves

Earthfault Protection (50N ANSI) - EFIPTOC

	Unit	Step	Value
Start Value:	i/pu	0,01	0,10-40,00
Operate Delay Time:	s	0,01	0,02-200,00
Operation:	Off / On	-	On
Start Value Mult (...*In):	i/pu		1,57
Time Delay:	s		0,5
Operating Curve Type:			DT (IEC Def. time)

Earthfault Protection (51N-1 ANSI) - EFLPTOC

	Unit	Step	Value
Start Value:	i/pu	0,005	0,010-5,000
Operate Delay Time:	s	0,01	0,04-200,00
Operation:	Off / On	-	On
Start Value Mult (...*In):	i/pu		0,10
Time Delay:	s		6,00
Operating Curve Type:			DT (IEC Def. time)

Circuit breaker failure protection (50BF ANSI) - CCBRRBF

	Unit	Step	Value
Operation:	Off / On	-	On
Current Value (range):	i/pu	0,05	0,05-1,00
Current value Res (range):	i/pu	0,05	0,05-1,00
Retrip time (range):	s	0,01	0,00-60,00
CB failure delay (range):	s	0,01	0,00-60,00
Current Value:	i/pu	0,05	0,30
Current value Res:	i/pu	0,05	0,30
Retrip time:	s	0,01	0
CB failure delay:	s	0,01	0,15

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Auto - recloser (79 ANSI) - DARREC

	Unit	Step	Value
Operation:	Off / On	-	On
Reclosing operation:	Off External Ctl On	-	External Ctl
Terminal priority:	None/ Low (follower)/ High (master)	-	None
Synchronisation set:	-	1	0
Auto initiation Cnd:	not allowed when sync fails/ CB doesn't close/ Both	-	when sync fails
Init signals CBB1:	-	1	1
Init signals CBB2:	-	1	0
Init signals CBB3:	-	1	0
Init signals CBB4:	-	1	0
Init signals CBB5:	-	1	0
Init signals CBB6:	-	1	0
Init signals CBB7:	-	1	0
Frq Op counter limit:	-	1	0
Reclaim time:	s	0.1	10
First reclose time:	s	0.01	3
Second reclose time:	s	0.01	5
Third reclose time:	s	0.01	5
Fourth reclose time:	s	0.01	5
Fifth reclose time:	s	0.01	5
Sixth reclose time:	s	0.01	5
Seventh reclose time:	s	0.01	5
Frq Op counter time:	Min	1	1
Frq Op recovery time:	Min	1	1
Close pulse time:	s	0.01	0,2
Disable Pro signal:	-	1	63
Auto init:	-	1	0
Fourth delay in SOFT:	Not in Use In use	-	Not In Use

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IED Type: REF630(INCOMER - =AA1J00+J03)

This IED provides overcurrent & earth fault protection for the MV network. The aforementioned IED also provides over & undervoltage, over & under frequency, AC reclosing, breaker failure and frequency gradient functions. The basic IED function settings are illustrated in the latter tables:

Overload Protection (51P-1 ANSI) - PHLPTOC

	Unit	Step	Value
Start Value:	i/pu	0,01	0,05-5,0
Time Multiplier (range):	t/pu	0,05	0,05-15,00
Operation:	Off / On	-	On
Num of start phases:	1 out of 3 2 out of 3 3 out of 3	-	1 out of 3
Start Value Mult (...*In):	i/pu		0,95
Time Multiplier:	t/pu		1,00
Operating Curve Type:			IDMT (IEC normal inverse)

Overcurrent Protection (51P-2 ANSI) - PHHPTOC

	Unit	Step	Value
Start Value:	i/pu	0,01	0,10-40,00
Operate Delay Time:	s	0,01	0,02-200,00
Operation:	Off / On	-	On
Num of start phases:	1 out of 3 2 out of 3 3 out of 3	-	1 out of 3
Start Value Mult (...*In):	i/pu		2,29
Time Delay:	s		0,50
Operating Curve Type:			DT (IEC Def. time)

Earthfault Protection (51N-2 ANSI) - EFHPTOC

	Unit	Step	Value
Start Value:	i/pu	0,01	0,10-40,00
Time Multiplier (range):	t/pu	0,05	0,05-15,00
Operation:	Off / On	-	On
Start Value Mult (...*In):	i/pu		0,9
Time Multiplier:	t/pu		0,3
Operating Curve Type:			IDMT (IEC normal inverse)
Operate Delay Time:	s	0,01	0,02-200,00
Time Delay:	s		Applicable only for DT curves

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Earthfault Protection (50N ANSI) - EFIPTOC

	Unit	Step	Value
Start Value:	i/pu	0,01	0,10-40,00
Operate Delay Time:	s	0,01	0,02-200,00
Operation:	Off / On	-	On
Start Value Mult (...*In):	i/pu		1,95
Time Delay:	s		0,50
Operating Curve Type:			DT (IEC Def. time)

Earthfault Protection (51N-1 ANSI) - EFLPTOC

	Unit	Step	Value
Start Value:	i/pu	0,005	0,010-5,000
Operate Delay Time:	s	0,01	0,04-200,00
Operation:	Off / On	-	On
Start Value Mult (...*In):	i/pu		0,08
Time Delay:	s		5,05
Operating Curve Type:			DT (IEC Def. time)

Overfrequency Protection (81O ANSI) - DAPTOF

	Unit	Step	Value
Start Value:	Hz	0,1	35,0-64,0
Operate Delay Time:	s	0,001	0,170-60,000
Start Value Mult (...*fn):	Hz		1,04*fn = 52
Delay Time:	s		0,2
Operation:	Off / On	-	On
Operating Curve Type:			DT (IEC Def. time)

Underfrequency Protection (81O ANSI) - DAPTUF

	Unit	Step	Value
Start Value:	Hz	0,1	35,0-64,0
Operate Delay Time:	s	0,001	0,170-60,000
Start Value Mult (...*fn):	Hz		0,94*fn = 47
Delay Time:	s		0,2
Operation:	Off / On	-	On
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Circuit breaker failure protection (50BF ANSI) - CCBRRBF

	Unit	Step	Value
Operation:	Off / On	-	On
Current Value (range):	i/pu	0,05	0,05-1,00
Current value Res (range):	i/pu	0,05	0,05-1,00
Retrip time (range):	s	0,01	0,00-60,00
CB failure delay (range):	s	0,01	0,00-60,00
Current Value:	i/pu	0,05	1,00
Current value Res:	i/pu	0,05	1,00
Retrip time:	s	0,01	0
CB failure delay:	s	0,01	0,2

Three - Phase Overvoltage Protection (59 ANSI) - 3U> PHPTOV

	Unit	Step	Value
Operation:	Off / On	-	On
Start Value:	u/pu	0,01	0,05-1,6
Time Multiplier (range):	t/pu	0,05	0,05-15,00 (Only in use when IDMT (IEC) curves are used)
Operate Delay Time:	s	0,01	0,04-300,00
Voltage selection:	phase-to-earth phase-to-phase	-	phase-to-phase
Num of start phases:	1 out of 3 2 out of 3 3 out of 3	-	1 out of 3
Operating Curve Type:	ANSI Def. Time IEC Def. Time Inv. Curve A Inv. Curve B Inv. Curve C Programmable	-	DT (IEC Def. time)
Start Value (setting):	u/pu		1,08
Time Delay:	s		90

Prepared: YPg 11-03-2015	Responsible Department: KEP	Doc.Kind: Calculation study	DCC: & &EED
Approved: APr 11-03-2015	Take Over Department: -	Title: Selectivity Study & Relay Setting Calculation	Language: en
	Derived from: - Replaces: -	Item: -	
 ABB SA		Doc.No: 3VGR111416E0002	Rev.: A
			Sh/No.Sh 13/27

Three - Phase Overvoltage Protection (59 ANSI) - 3U>> PHPTOV

	Unit	Step	Value
Operation:	Off / On	-	On
Start Value:	u/pu	0,01	0,05-1,6
Time Multiplier (range):	t/pu	0,05	0,05-15,00 (Only in use when IDMT (IEC) curves are used)
Operate Delay Time:	s	0,01	0,04-300,00
Voltage selection:	phase-to-earth phase-to-phase	-	phase-to-phase
Num of start phases:	1 out of 3 2 out of 3 3 out of 3	-	1 out of 3
Operating Curve Type:	ANSI Def. Time IEC Def. Time Inv. Curve A Inv. Curve B Inv. Curve C Programmable	-	DT (IEC Def. time)
Start Value (setting):	u/pu		1,15
Time Delay:	s		3

Three - Phase Overvoltage Protection (59 ANSI) - 3U>>> PHPTOV

	Unit	Step	Value
Operation:	Off / On	-	On
Start Value:	u/pu	0,01	0,05-1,6
Time Multiplier (range):	t/pu	0,05	0,05-15,00 (Only in use when IDMT (IEC) curves are used)
Operate Delay Time:	s	0,01	0,04-300,00
Voltage selection:	phase-to-earth phase-to-phase	-	phase-to-phase
Num of start phases:	1 out of 3 2 out of 3 3 out of 3	-	1 out of 3
Operating Curve Type:	ANSI Def. Time IEC Def. Time Inv. Curve A Inv. Curve B Inv. Curve C Programmable	-	DT (IEC Def. time)
Start Value (setting):	u/pu		1,22
Time Delay:	s		0,2

Prepared: YPg 11-03-2015	Responsible Department: KEP	Doc.Kind: Calculation study	DCC: & &EED
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	Derived from: - Replaces: -	Item: -	
 ABB SA		Doc.No: 3VGR111416E0002	Rev.: A
			Sh/No.Sh 14/27

Three - Phase Uvervoltage Protection (27 ANSI) - PHPTUV

	Unit	Step	Value
Operation:	Off / On	-	On
Start Value:	u/pu	0,01	0,05-1,2
Time Multiplier (range):	t/pu	0,05	0,05-15,00 (Only in use when IDMT curves are used)
Operate Delay Time:	s	0,01	0,04-300,00
Voltage selection:	phase-to-earth phase-to-phase	-	phase-to-phase
Num of start phases:	1 out of 3 2 out of 3 3 out of 3	-	1 out of 3
Phase supervision:	-	-	All
Operating Curve Type:	ANSI Def. Time/ IEC Def. Time/ Inv. Curve A/ Inv. Curve B/Programmable	-	DT (IEC Def. time)
Start Value (setting):	u/pu		0,80
Time Delay:	s		2,5

Prepared: YPg 11-03-2015	Responsible Department: KEP	Doc.Kind: Calculation study	DCC: & &EED
Approved: APr 11-03-2015	Take Over Department: -	Title: Selectivity Study & Relay Setting Calculation	Language: en
	Derived from: - Replaces: -	Item: -	
 ABB SA		Doc.No: 3VGR111416E0002	Rev.: A
			Sh/No.Sh 15/27

Auto - recloser (79 ANSI) - DARREC

	Unit	Step	Value
Operation:	Off / On	-	On
Reclosing operation:	Off External Ctl On	-	External Ctl
Terminal priority:	None/ Low (follower)/ High (master)	-	None
Synchronisation set:	-	1	0
Auto initiation Cnd:	not allowed when sync fails/ CB doesn't close/ Both	-	when sync fails
Init signals CBB1:	-	1	1
Init signals CBB2:	-	1	0
Init signals CBB3:	-	1	0
Init signals CBB4:	-	1	0
Init signals CBB5:	-	1	0
Init signals CBB6:	-	1	0
Init signals CBB7:	-	1	0
Frq Op counter limit:	-	1	0
Reclaim time:	s	0.1	10
First reclose time:	s	0.01	5
Second reclose time:	s	0.01	5
Third reclose time:	s	0.01	5
Fourth reclose time:	s	0.01	5
Fifth reclose time:	s	0.01	5
Sixth reclose time:	s	0.01	5
Seventh reclose time:	s	0.01	5
Frq Op counter time:	Min	1	1
Frq Op recovery time:	Min	1	1
Close pulse time:	s	0.01	0,2
Disable Pro signal:	-	1	63
Auto init:	-	1	0
Fourth delay in SOFT:	Not in Use In use	-	Not In Use

Prepared: YPg 11-03-2015	Responsible Department: KEP	Doc.Kind: Calculation study	DCC: & &EED
Approved: APr 11-03-2015	Take Over Department: -	Title: Selectivity Study & Relay Setting Calculation	Language: en
	Derived from: - Replaces: -	Item: -	
 ABB SA		Doc.No: 3VGR111416E0002	Rev.: A
			Sh/No.Sh 16/27

IED Type: REF630 (=AA1J00+J01) (Underground Feeder 1)

This IED provides directional (towards the wind farm loop 2) and non-directional overcurrent & earth fault protection to feeder panel 1 feeding cables, the MV network and to the wind farm (loop 2). This IED also provides breaker failure protection function. The abovementioned IEDs are situated in the feeder panel 1 within the TP Aeolian Dynamics S/S. The settings recommended are:

Overload Protection (51P-1 ANSI) - PHLPTOC

	Unit	Step	Value
Start Value:	i/pu	0,01	0,05-5,0
Time Multiplier (range):	t/pu	0,05	0,05-15,00
Operation:	Off / On	-	On
Num of start phases:	1 out of 3 2 out of 3 3 out of 3	-	1 out of 3
Start Value Mult (...*In):	i/pu		0,9
Time Multiplier:	t/pu		0,15
Operating Curve Type:			LI (IEC long inverse)

Overcurrent Protection (51P-2 ANSI) - PHHPTOC

	Unit	Step	Value
Start Value:	i/pu	0,01	0,10-40,00
Operate Delay Time:	s	0,01	0,02-200,00
Operation:	Off / On	-	On
Num of start phases:	1 out of 3 2 out of 3 3 out of 3	-	1 out of 3
Start Value Mult (...*In):	i/pu		3,29
Time Delay:	s		0,30
Operating Curve Type:			DT (IEC Def. time)

Directional Overcurrent Protection (67-2 ANSI) - DPHHPDOC

	Unit	Step	Value
Start Value:	i/pu	0,01	0,10-40,00
Operate Delay Time:	s	0,01	0,04-200
Operation:	Off / On	-	On
Num of start phases:	1 out of 3 2 out of 3 3 out of 3	-	1 out of 3
Start Value Mult (...*In):	i/pu		2,61
Time Delay:	s		0,3
Operating Curve Type:			DT (IEC Def. time)
Directional Mode:			Forward
Characteristic Angle:	Deg	1	0

Prepared: YPg 11-03-2015	Responsible Department: KEP	Doc.Kind: Calculation study	DCC: & &EED
Approved: APr 11-03-2015	Take Over Department: -	Title: Selectivity Study & Relay Setting Calculation	Language: en
	Derived from: - Replaces: -	Item: -	
 ABB SA		Doc.No: 3VGR111416E0002	Rev.: A
			Sh/No.Sh 17/27

Earthfault Protection (51N-2 ANSI) - EFHPTOC

	Unit	Step	Value
Start Value:	i/pu	0,01	0,10-40,00
Time Multiplier (range):	t/pu	0,05	0,05-15,00
Operation:	Off / On	-	On
Start Value Mult (...*In):	i/pu		1,21
Time Multiplier:	t/pu		0,15
Operating Curve Type:			IDMT (IEC normal inverse)
Operate Delay Time:	s	0,01	0,02-200,00
Time Delay:	s		Applicable only for DT curves

Earthfault Protection (50N ANSI) - EFIPTOC

	Unit	Step	Value
Start Value:	i/pu	0,01	0,10-40,00
Operate Delay Time:	s	0,01	0,02-200,00
Operation:	Off / On	-	On
Start Value Mult (...*In):	i/pu		2,89
Time Delay:	s		0,3
Operating Curve Type:			DT (IEC Def. time)

Earthfault Protection (67N-2 ANSI) - DEFHPDEF

	Unit	Step	Value
Operation:	Off / On	-	On
Directional Mode:			Forward
Start Value:	i/pu	0,01	0,1-40,0
Voltage Start Value (range):	pu	0,001	0,01-1,00
Voltage Start Value:	pu	0,001	0,01
Characteristic Angle:	Deg	1	0
Start Value Mult (...*In):	i/pu		2,89
Operating Curve Type:			DT (IEC Def. time)
Operate Delay Time:	s	0,01	0,06-200,00
Time Delay:	s		0,3

Prepared: YPg 11-03-2015	Responsible Department: KEP	Doc.Kind: Calculation study	DCC: & &EED
Approved: APr 11-03-2015	Take Over Department: -	Title: Selectivity Study & Relay Setting Calculation	Language: en
	Derived from: - Replaces: -	Item: -	
 ABB SA		Doc.No: 3VGR111416E0002	Rev.: A
			Sh/No.Sh 18/27

Circuit breaker failure protection (50BF ANSI) - CCBRRBF

	Unit	Step	Value
Operation:	Off / On	-	On
Current Value (range):	i/pu	0,05	0,05-1,00
Current value Res (range):	i/pu	0,05	0,05-1,00
Retrip time (range):	s	0,01	0,00-60,00
CB failure delay (range):	s	0,01	0,00-60,00
Current Value:	i/pu	0,05	0,30
Current value Res:	i/pu	0,05	0,30
Retrip time:	s	0,01	0
CB failure delay:	s	0,01	0,15

IED Type: REF630 (=AA1J00+J02) (Underground Feeder 2)

This IED provides directional (towards the wind farm loop 1) and non-directional overcurrent & earth fault protection to feeder panel 2 feeding cables, the MV network and to the wind farm (loop 1). This IED also provides breaker failure protection function. The abovementioned IEDs are situated in the feeder panel 2 within the TP Aeolian Dynamics S/S. The settings recommended are:

Overload Protection (51P-1 ANSI) - PHLPTOC

	Unit	Step	Value
Start Value:	i/pu	0,01	0,05-5,0
Time Multiplier (range):	t/pu	0,05	0,05-15,00
Operation:	Off / On	-	On
Num of start phases:	1 out of 3 2 out of 3 3 out of 3	-	1 out of 3
Start Value Mult (...*In):	i/pu		0,9
Time Multiplier:	t/pu		0,15
Operating Curve Type:			LI (IEC long inverse)

Overcurrent Protection (51P-2 ANSI) - PHHPTOC

	Unit	Step	Value
Start Value:	i/pu	0,01	0,10-40,00
Operate Delay Time:	s	0,01	0,02-200,00
Operation:	Off / On	-	On
Num of start phases:	1 out of 3 2 out of 3 3 out of 3	-	1 out of 3
Start Value Mult (...*In):	i/pu		3,29
Time Delay:	s		0,30
Operating Curve Type:			DT (IEC Def. time)

Prepared: YPg 11-03-2015	Responsible Department: KEP	Doc.Kind: Calculation study	DCC: & &EED
Approved: APr 11-03-2015	Take Over Department: -	Title: Selectivity Study & Relay Setting Calculation	Language: en
	Derived from: - Replaces: -	Item: -	
 ABB SA		Doc.No: 3VGR111416E0002	Rev.: A
			Sh/No.Sh 19/27

Directional Overcurrent Protection (67-2 ANSI) - DPHHPDOC

	Unit	Step	Value
Start Value:	i/pu	0,01	0,10-40,00
Operate Delay Time:	s	0,01	0,04-200
Operation:	Off / On	-	On
Num of start phases:	1 out of 3 2 out of 3 3 out of 3	-	1 out of 3
Start Value Mult (...*In):	i/pu		2,61
Time Delay:	s		0,3
Operating Curve Type:			DT (IEC Def. time)
Directional Mode:			Forward
Characetristic Angle:	Deg	1	0

Earthfault Protection (51N-2 ANSI) - EFHPTOC

	Unit	Step	Value
Start Value:	i/pu	0,01	0,10-40,00
Time Multiplier (range):	t/pu	0,05	0,05-15,00
Operation:	Off / On	-	On
Start Value Mult (...*In):	i/pu		1,21
Time Multiplier:	t/pu		0,15
Operating Curve Type:			IDMT (IEC normal inverse)
Operate Delay Time:	s	0,01	0,02-200,00
Time Delay:	s		Applicable only for DT curves

Earthfault Protection (50N ANSI) - EFIPTOC

	Unit	Step	Value
Start Value:	i/pu	0,01	0,10-40,00
Operate Delay Time:	s	0,01	0,02-200,00
Operation:	Off / On	-	On
Start Value Mult (...*In):	i/pu		2,89
Time Delay:	s		0,3
Operating Curve Type:			DT (IEC Def. time)

Prepared: YPg 11-03-2015	Responsible Department: KEP	Doc.Kind: Calculation study	DCC: & &EED
Approved: APr 11-03-2015	Take Over Department: -	Title: Selectivity Study & Relay Setting Calculation	Language: en
	Derived from: - Replaces: -	Item: -	
 ABB SA		Doc.No: 3VGR111416E0002	Rev.: A
			Sh/No.Sh 20/27

Earthfault Protection (67N-2 ANSI) - DEFHPDEF

	Unit	Step	Value
Operation:	Off / On	-	On
Directional Mode:			Forward
Start Value:	i/pu	0,01	0,1-40,0
Voltage Start Value (range):	pu	0,001	0,01-1,00
Voltage Start Value:	pu	0,001	0,01
Characetristic Angle:	Deg	1	0
Start Value Mult (... *In):	i/pu		2,89
Operating Curve Type:			DT (IEC Def. time)
Operate Delay Time:	s	0,01	0,06-200,00
Time Delay:	s		0,3

Circuit breaker failure protection (50BF ANSI) - CCBRRBF

	Unit	Step	Value
Operation:	Off / On	-	On
Current Value (range):	i/pu	0,05	0,05-1,00
Current value Res (range):	i/pu	0,05	0,05-1,00
Retrip time (range):	s	0,01	0,00-60,00
CB failure delay (range):	s	0,01	0,00-60,00
Current Value:	i/pu	0,05	0,30
Current value Res:	i/pu	0,05	0,30
Retrip time:	s	0,01	0
CB failure delay:	s	0,01	0,15

Prepared: YPg 11-03-2015	Responsible Department: KEP	Doc.Kind: Calculation study	DCC: & &EED
Approved: APr 11-03-2015	Take Over Department: -	Title: Selectivity Study & Relay Setting Calculation	Language: en
	Derived from: - Replaces: -	Item: -	
 ABB SA		Doc.No: 3VGR111416E0002	Rev.: A
			Sh/No.Sh 21/27

IED Type: ecorRPG – 202A

This IED is used for the protection of the MV network, the wind generator MV board feeding cable and the network of the interconnecting cables, between wind generators. The abovementioned IED incorporates overload, overcurrent & earth fault protection. The abovementioned IED is placed within the MV panel of each wind generator. The settings recommended for the 50/51 and 51N/50N protections are:

- Phase full load current

Pt :

Tr :

In ...A

Installation Transformer power / (W)

Line voltage / (V)

$$I_n = \frac{P_t}{((T_r \times \sqrt{3}) \times \cos \phi)} = \frac{2.1 \cdot 10^6}{(22 \cdot 10^3 \cdot \sqrt{3}) \times 0.9} \approx 61(A) \text{ A}$$

Current Sensor Ratio :

1000 / 1

Method for entering parameters :

manual

- Phase Protection - 51

Overload Factor :

1.1* In = 67.1 A

Curve Type :

VI

Constant Multiplier :

0.2

- Phase Protection - 50

Short-circuit Factor :

7,0* In = 427 A

Trip Time :

100 [ms]

Tripping Enabled :

DT

- Earth Protection - 51N

Earth Leakage Factor :

0.1* In = 6.1 A

Curve Time :

DT

Constant Multiplier :

0.05

- Earth Protection - 50N

Short-circuit Factor :

2* In = 122 A

Trip Time :

100 [ms]

Tripping Enabled :

DT

For more detailed settings see Annex B2.

Prepared: YPg 11-03-2015	Responsible Department: KEP	Doc.Kind: Calculation study	DCC: & &EED
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 ABB SA		Doc.No: 3VGR111416E0002	Rev.: A
			Sh/No.Sh 22/27

5. Annex A

The data presented in this paragraph were elicited from the reference documents, presented above.

Prepared: YPg 11-03-2015	Responsible Department: KEP	Doc.Kind: Calculation study	DCC: & &EED
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 ABB SA		Doc.No: 3VGR111416E0002	Rev.: A
			Sh/No.Sh 23/27

NETWORK FEEDER (EAC NETWORK)

NAME	Sk"min	Sk"max	Ik"min	Ik"max	Z(0)/Z(1)min	Z(0)/Z(1)max	R(1)/X(1)min	R(1)/X(1)max	R(0)/X(0)min	R(0)/X(0)max
	MVA	MVA	kA	kA						
EAC NETWORK	80,783	163	2,12	4,278	1,631	2,9846	0,1	0,1	0,1	0,1

LINE DATA

NAME	TYPE	LENGTH	Number of parallel cables	R(1)	X(1)	C(1)	R(0)	X(0)	C(0)	Q	Material	Temp. end of SC	Max Oper.T emp.	Reduction factor
		km		Ohm/km	Ohm/km	μF/km	Ohm/km	Ohm/km	μF/km	mm ²		°C	°C	
3x(1x300mm ²)	N2XSY 3X1X300 12/22kV	0,1	1	0,0128	0,011	0,325	0,0685	0,0275	0,325	300	Al	250	90	0,809
Cable C1	N2XSY 3x(1x150mm ²) 12/22kV	0,706	1	0,269	0,00039	0,249	1,54944	0,00109624	0,249	150	Al	250	90	0,688
Cable C2	N2XSY 3x(1x150mm ²) 12/22kV	0,77	1	0,269	0,00039	0,249	1,54944	0,00109624	0,249	150	Al	250	90	0,593
Cable C3	N2XSY 3x(1x150mm ²) 12/22kV	0,487	1	0,269	0,00039	0,249	1,54944	0,00109624	0,249	150	Al	250	90	0,688
Cable C4	N2XSY 3x(1x150mm ²) 12/22kV	0,346	1	0,269	0,00039	0,249	1,54944	0,00109624	0,249	150	Al	250	90	0,593
Cable C5	N2XSY 3x(1x150mm ²) 12/22kV	0,994	1	0,269	0,00039	0,249	1,54944	0,00109624	0,249	150	Al	250	90	0,593
Cable C6	N2XSY 3x(1x150mm ²) 12/22kV	0,542	1	0,269	0,00039	0,249	1,54944	0,00109624	0,249	150	Al	250	90	0,688
CBL WG_1	EPR 3x70/70mm ² 22kV	0,095	1	0,347	0,099	0,325	1,041	0,297	0,325	70	Cu	250	120	0,473
CBL WG_2	EPR 3x70/70mm ² 22kV	0,095	1	0,347	0,099	0,325	1,041	0,297	0,325	70	Cu	250	120	0,473
CBL WG_3	EPR 3x70/70mm ² 22kV	0,095	1	0,347	0,099	0,325	1,041	0,297	0,325	70	Cu	250	120	0,473
CBL WG_4	EPR 3x70/70mm ² 22kV	0,095	1	0,347	0,099	0,325	1,041	0,297	0,325	70	Cu	250	120	0,473
CBL WG_5	EPR 3x70/70mm ² 22kV	0,095	1	0,347	0,099	0,325	1,041	0,297	0,325	70	Cu	250	120	0,473
CBL WG_6	EPR 3x70/70mm ² 22kV	0,095	1	0,347	0,099	0,325	1,041	0,297	0,325	70	Cu	250	120	0,473

TRANSFORMERS DATA

NAME	Vector	Sr	Ur1	Ur2	ukr(1)	uRr(1)	ukr(0)	uRr(0)	I0	Pfe	Earthing at primary	Earthing at secondary
		MVA	kV	kV	%	%	%	%	%	kW		
TR (WG 1)	DYn5	2,1	22	0,69	7,6	0,1857	0,594	0,1857	0	3,9	na	direct
TR (WG 2)	DYn5	2,1	22	0,69	7,6	0,1857	0,594	0,1857	0	3,9	na	direct
TR (WG 3)	DYn5	2,1	22	0,69	7,6	0,1857	0,594	0,1857	0	3,9	na	direct
TR (WG 4)	DYn5	2,1	22	0,69	7,6	0,1857	0,594	0,1857	0	3,9	na	direct
TR (WG 5)	DYn5	2,1	22	0,69	7,6	0,1857	0,594	0,1857	0	3,9	na	direct
TR (WG 6)	DYn5	2,1	22	0,69	7,6	0,1857	0,594	0,1857	0	3,9	na	direct

ASYNCHRONOUS DOUBLE FED WIND GENERATORS

NAME	Pr	Sr	Ur	Ir	cosphi	eta	Ia/Ir	pole pairs	cosphi start	Ma/Mr	Mk/Mr	Rm	sr	Slip
	MW	MVA	kV	kA	-	-	-	-	-	-	-	Ohm	%	%
WG (A-1)	1,8	2,062	0,69	1,7252	0,9	0,97	6,474	2	0,2	0,2917	3,0722	0,007173	0,6	-12
WG (A-2)	1,8	2,062	0,69	1,7252	0,9	0,97	6,474	2	0,2	0,2917	3,0722	0,007173	0,6	-12
WG (A-3)	1,8	2,062	0,69	1,7252	0,9	0,97	6,474	2	0,2	0,2917	3,0722	0,007173	0,6	-12
WG (A-4)	1,8	2,062	0,69	1,7252	0,9	0,97	6,474	2	0,2	0,2917	3,0722	0,007173	0,6	-12
WG (A-5)	1,8	2,062	0,69	1,7252	0,9	0,97	6,474	2	0,2	0,2917	3,0722	0,007173	0,6	-12
WG (A-6)	1,8	2,062	0,69	1,7252	0,9	0,97	6,474	2	0,2	0,2917	3,0722	0,007173	0,6	-12

6. Annex B1

Load Flow

L.F Type	At 22kV
Maximum Load Current (A) - (REF 630 [INCOMER])	605,8
Maximum Load Current (A) - (REF 630 - [IPPM 1])	488,1
Maximum Load Current (A) - (REF 630 - [IPPM 2])	488,1
Maximum Load Current (A) - (ecorRPG - 202A)	67,1

Fault Levels (To be used for selectivity)

S.C Type	Value
Minimum S.C 1- phase (kA) (REF 630 [INCOMER])	1,747
Minimum S.C 2- phase (kA) (REF 630 [INCOMER])	1,834
Minimum S.C 1- phase (kA) (REF 630 - [IPPM 1])	1,747
Minimum S.C 2- phase (kA) (REF 630 - [IPPM 1])	1,834
Minimum S.C 1- phase (kA) (REF 630 - [IPPM 2])	1,747
Minimum S.C 2- phase (kA) (REF 630 - [IPPM 2])	1,834
Minimum S.C 1- phase (kA) (ecorRPG - 202A)	1,578
Minimum S.C 2- phase (kA) (ecorRPG - 202A)	1,770
Maximum S.C 3- phase (kA) (REF 630 [INCOMER])	5,145
Maximum S.C 3- phase (kA) (REF 630 - [IPPM 1])	5,145
Maximum S.C 3- phase (kA) (REF 630 - [IPPM 2])	5,145
Maximum S.C 3- phase (kA) (ecorRPG - 202A)	5,101

Fault Current Contributions

S.C Type	Value
Minimum S.C 2- phase (kA) (REF 630 - [IPPM 1])	584
Minimum S.C 2- phase (kA) (REF 630 - [IPPM 2])	583

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7. Annex B2

ecorRPG - 202A Settings

Phase Protection (51 ANSI) - I>

Parameter	Meaning	Range	Step	Unit	Value
I> :	Phase curve type/ unit disabling	OFF,NI,VI,EI,DT	-	-	VI (IEC very inverse)
In.A:	Phase full load current	Manual or 192 or 480	-	A	61
I> :	Phase overload factor	1,00-1,30		i/pu	1,1
K :	Phase multiplier constant	0,05-1,6		t/pu	0,2

Phase Protection (50 ANSI) - I>>

Parameter	Meaning	Range	Step	Unit	Value
I>> :	Enabling instantaneous phase unit	OFF,DT	-	-	DT (IEC Def. time)
In.A:	Phase full load current	Manual or 192 or 480	-	A	61
I>> :	Phase instanataneous multiplier	1,0-25		i/pu	7,0
T>> :	Phase instantaneous time delay	0,05-2,5		t/pu	0,1

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Earth Protection (51N ANSI) - $I_0>$

Parameter	Meaning	Range	Step	Unit	Value
$I_0>$:	Zero-sequence curve type/ Unit disabling	OFF,NI,VI,EI,DT	-	-	NI (IEC Normal Inv. time)
In.A:	Phase full load current	Manual or 192 or 480	-	A	61
$I_0>$:	Earth leakage factor	0,1-0,8		i/pu	0,1
K_0 :	Zero-sequence multiplier constant	0,05-1,6		t/pu	0,05

Earth Protection (50N ANSI) - $I_0>>$

Parameter	Meaning	Range	Step	Unit	Value
$I_0>>$:	Enabling instantaneous earth unit	OFF,DT	-	-	DT (IEC Def. time)
In.A:	Phase full load current	Manual or 192 or 480	-	A	61
$I_0>>$:	Zero-sequence instantaneous multiplier	1,0-25		i/pu	2
$T_0>>$:	Zero-sequence instantaneous time delay	0,05-2,5		t/pu	0,1

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8. Annex C

The selectivity diagrams presented below, illustrate the IEDs adequacy.

Comments to figure 1:

In this figure it is illustrated that all IEDs allow the normal load flow through them without tripping. It should also be noted that the starting current of the wind generator has been taken into account in every scenario of the load flow calculation. The REF 630 of the incomer (=AA1J00+J03) is selective towards each underground feeder (=AA1J00+J01 & =AA1J00+J02) REF 630 protection element, while at the same time both are selective towards the ecorRPG-202A relay of the wind generator M.V. board. Furthermore the REF 630 of TP Aeolian Dynamics Feeder situated at Pseudas substation (=J04+J04), has the same settings as REF 630 of the incomer board (=AA1J00+J03) at TP Aeolian Dynamics substation and is selective towards all IEDs at TP Aeolian Dynamics substation.

Comments to figure 2:

In this figure are presented the non directional overcurrent diagrams depicting that the ecorRPG-202A element clears the minimum SC fault at the wind generator node quicker than the underground feeder (=AA1J00+J01 & =AA1J00+J02) and the incomer (=AA1J00+J03) protection elements. Accordingly the REF 630 "PHHPTOC" element of the feeder (=AA1J00+J01 & =AA1J00+J02) trips faster than the respective element of the incomer (=AA1J00+J03) at TP Aeolian Dynamics S/S and the TP Aeolian Dynamics feeder (=J04+J04) at Pseudas substation. The protection modules are set to trip at significantly lower thresholds so as to be activated by the individual contributions to the short circuit by each wind generator loop.

Comments to figure 3:

In this figure are presented the directional overcurrent diagrams depicting that the ecorRPG-202A element clears the minimum SC fault at the wind generator node quicker than the underground feeder (=AA1J00+J01 & =AA1J00+J02) "DPHHPDOC" and the incomer (=AA1J00+J03) "PHHPTOC" protection elements. Accordingly the REF 630 "PHHPTOC" element of the incomer (=AA1J00+J03) at TP Aeolian Dynamics S/S trips at the same time as the respective element of the TP Aeolian Dynamics feeder (=J04+J04) at Pseudas substation. The directional protection modules of the underground feeders (=AA1J00+J01 & =AA1J00+J02) at TP Aeolian Dynamics S/S are set to trip at significantly lower thresholds so as to be activated by the individual contributions to the short circuit by each wind generator loop.

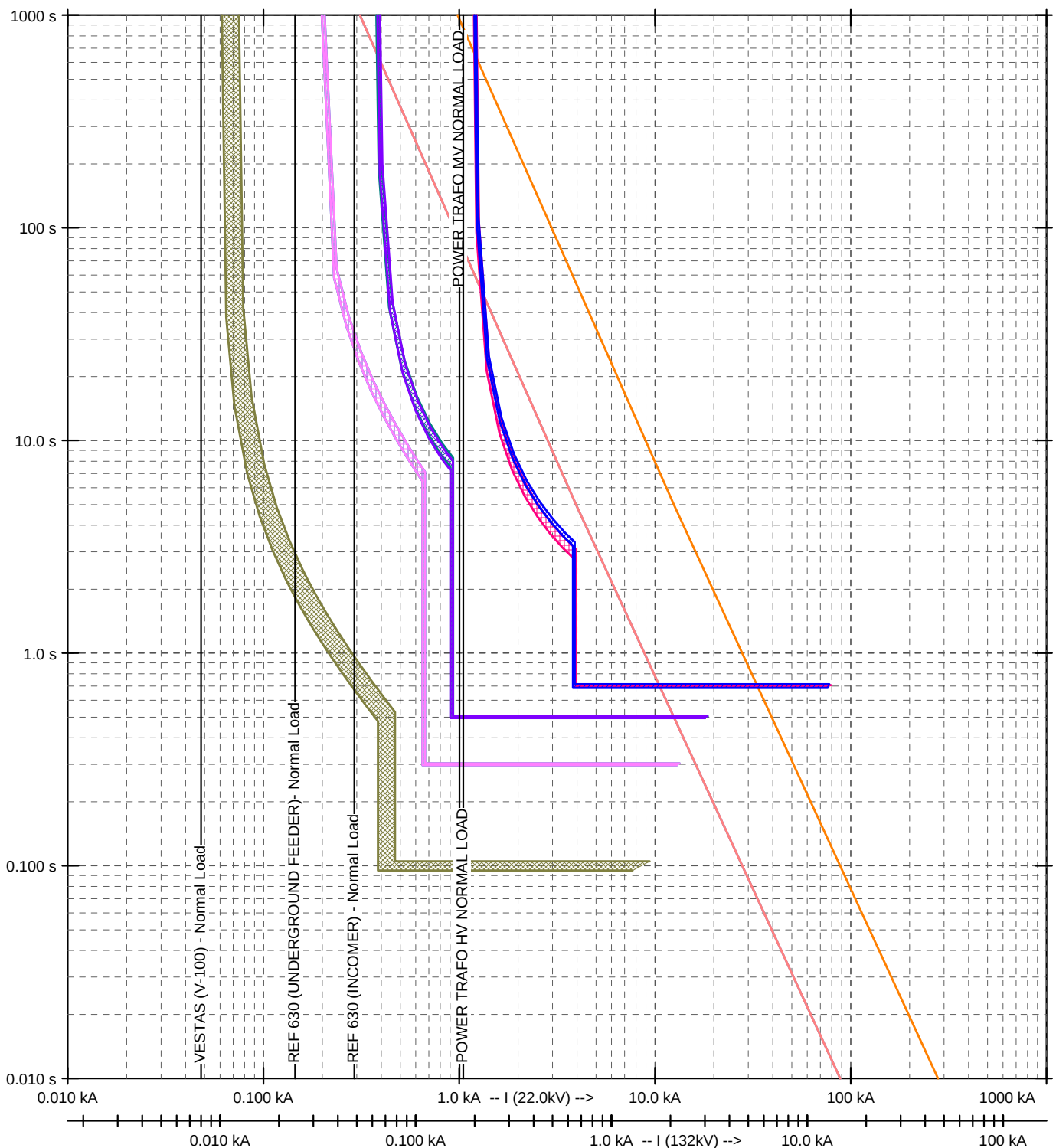
Comments to figure 4:

This figure illustrates the minimum earth fault thresholds of the wind farm network and the non directional elements of the IEDs used. The IEDs elements are selective to each other. The (51NS – "EFLPTOC") of the TP Aeolian Dynamics incomer (=AA1J00+J03) and TP Aeolian Dynamics Feeder (=J04+J04) at Pseudas have significantly lower thresholds than the (50N – "EFIPTOC") protection functions of each MV board respectively, but are time selective towards the other earth fault protection functions.

Comments to figure 5:

This figure illustrates the minimum earth fault thresholds of the wind farm network and the directional elements of the IEDs used. The IEDs elements are selective to each other. The (51NS – "EFLPTOC") of the TP Aeolian Dynamics incomer (=AA1J00+J03) and TP Aeolian Dynamics Feeder (=J04+J04) at Pseudas have significantly lower thresholds than the (50N – "EFIPTOC") protection functions of the incomer (=AA1J00+J03) and TP Aeolian Dynamics Feeder MV boards (=AA1J00+J01 & =AA1J00+J02) and the (67N – "DEFHPDEF") protection functions of the underground feeders (=AA1J00+J01 & =AA1J00+J02) respectively, but are time selective towards the other earth fault protection functions. The (67N – "DEFHPDEF") protection functions of the underground feeders (=AA1J00+J01 & =AA1J00+J02) have the same setting thresholds as the respective (50N) protection functions.

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-  ekorRPG-202A - TP AEOLIAN DYNAMICS WF : CGMCOSMOS_1 - CBL WG_1 (I>)
-  ORMAZABAL MV Panel (VESTAS IED)
-  CBL WG_6 : A-6 Trafo MV Side - CGMCOSMOS_6
-  Wind Generator cable between LV/MV Trafo and MV Panel ORMAZABAL
-  Cable C6 : WG_5 - WG_6
-  Wind Generator Interconnecting Cables
-  Cable_(3x1x300mm^2) : TP AEOLIAN DYNAMICS UGL - TP AEOLIAN DYNAMICS SS
-  REF_630 - TP AEOLIAN DYNAMICS SS (-Q1) : '=AA1J00+J01 - Underground Feeder (-Q1) (I>)
-  Feeder from Loop 1 of TP Aeolian Dynamics WF
-  REF_630 - TP AEOLIAN DYNAMICS SS (-Q2) : '=AA1J00+J02 - Underground Feeder (-Q2) (I>)
-  Feeder from Loop 2 of TP Aeolian Dynamics WF
-  REF_630 - TP AEOLIAN DYNAMICS SS (-Q3) : '=AA1J00+J03 - Incomer TP Aeolian (-Q3) (I>)
-  Incomer from 3x(1x300mm^2) underground cable of OHL TP Aeolian
-  REF_630 - PSEVDAS SS_IPPM : '=J04+J04 - TP Aeolian Feeder (I>)
-  7X1.2 MV Feeder Panel to which an OHL of Al 70mm^2 connects
-  REF_630 - PSEVDAS SS_TM : '=J03+J03 - Incomer No1 (I>)
-  Incomer from MV E06 Bay of Psevdas SS protecting Trafo Cables
- REC_670 - PSEVDAS SS_HV_PP : '=E06+CRP06 - Transformer Feeder (I>)
- HV Control & Protection Panel Bay E06

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D-68128 Mannheim

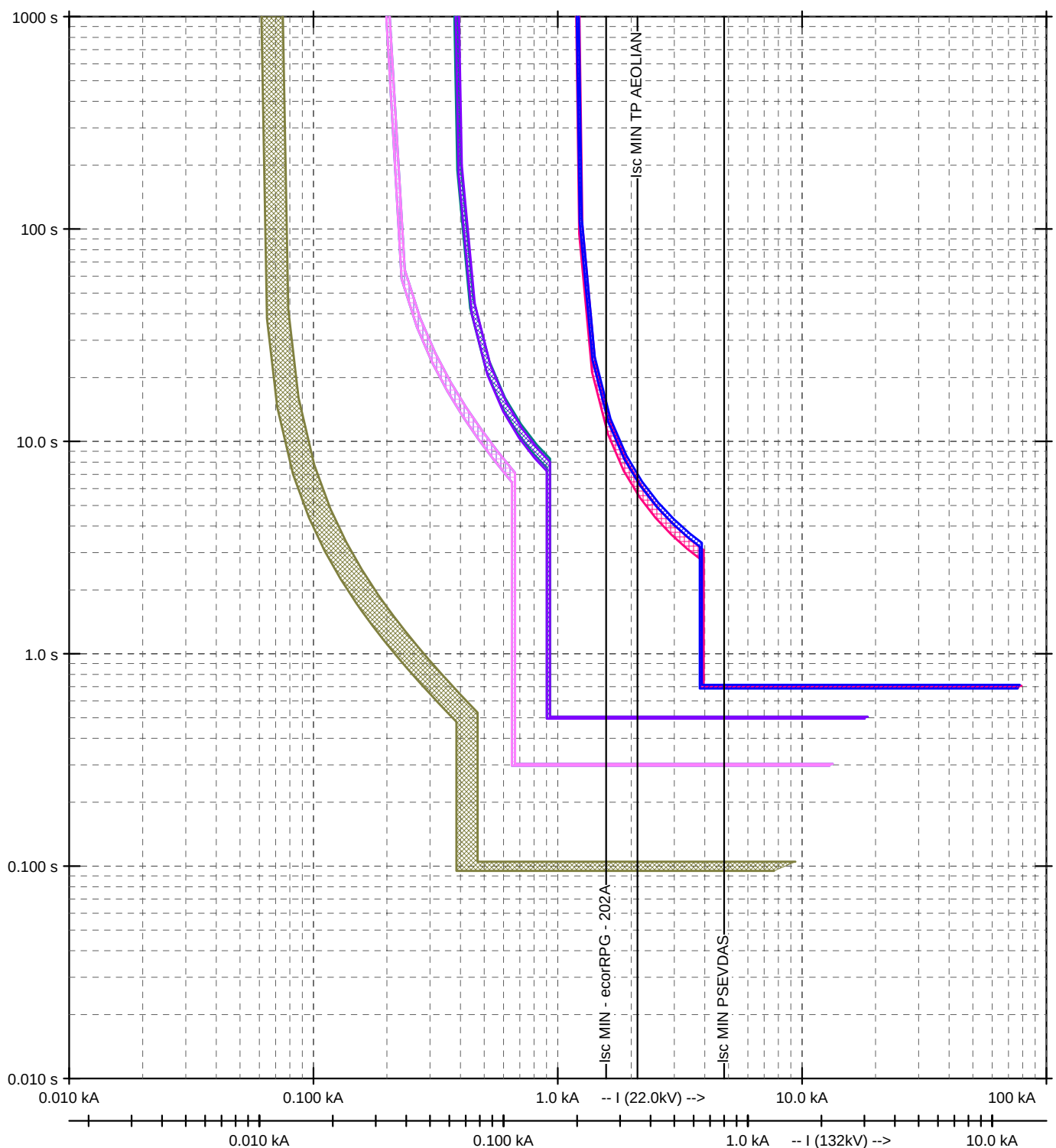
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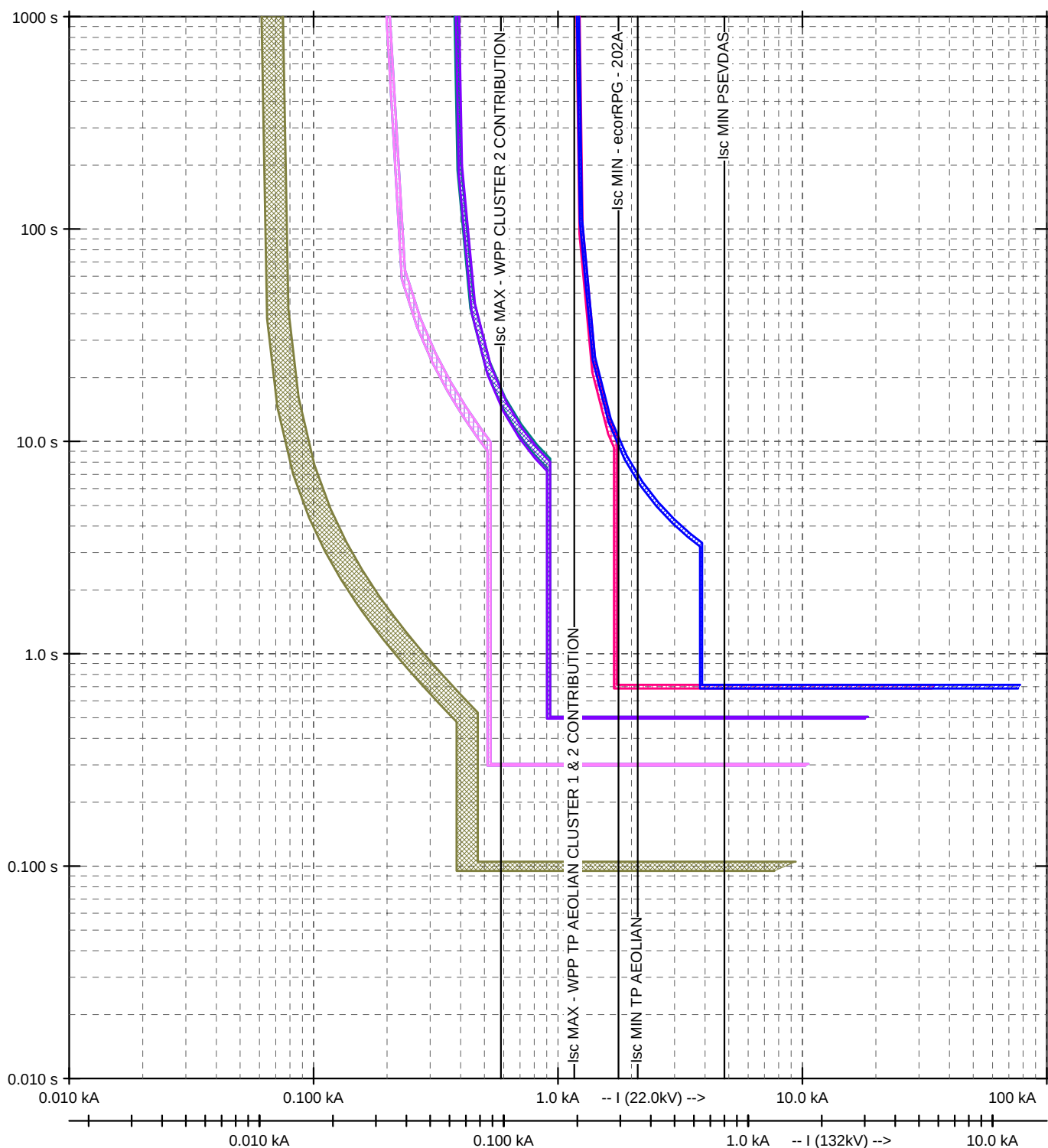
Overload Protection - 51 (ANSI)

Fig.No : 3VGR111416E0002_Fig_1

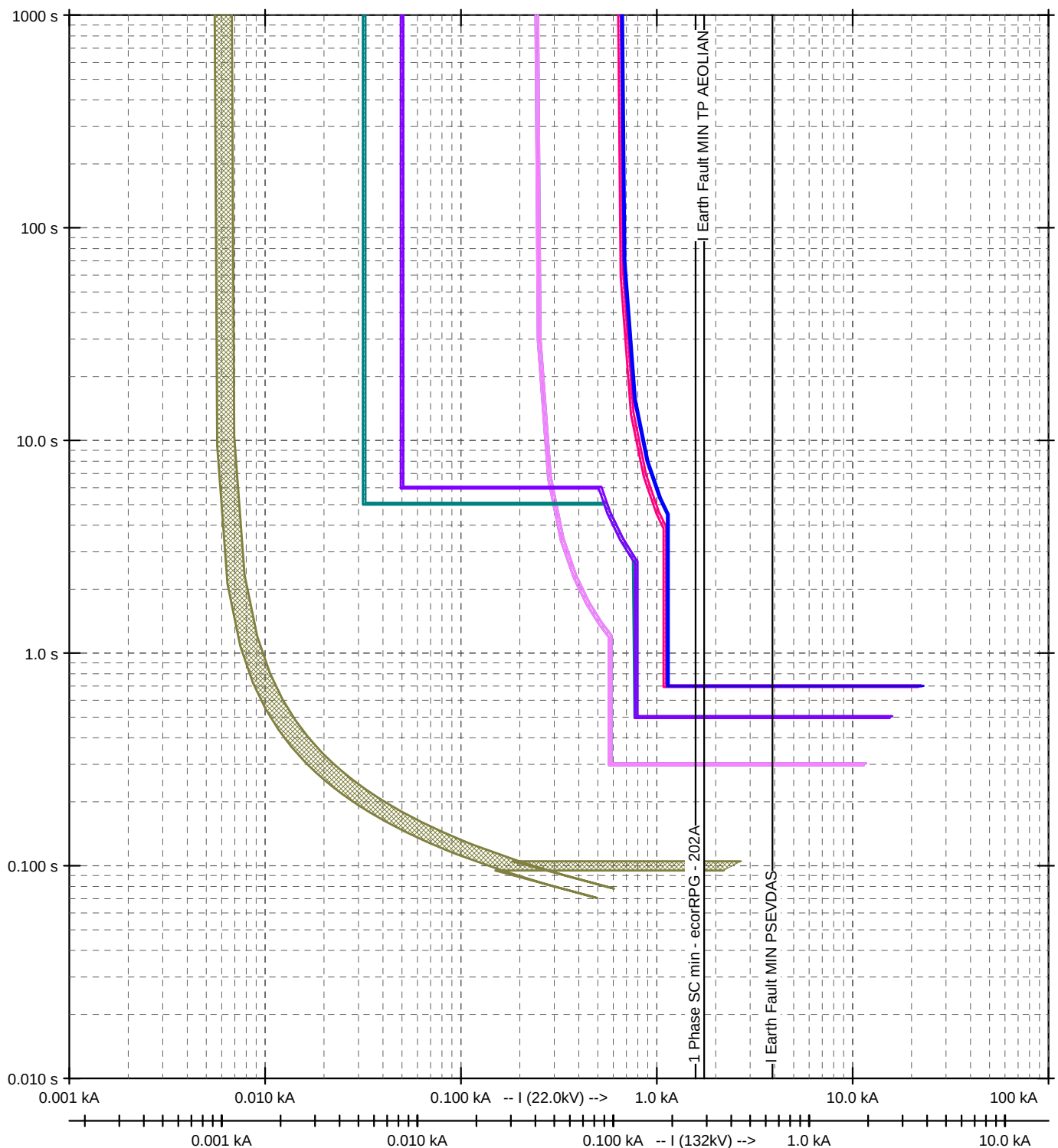
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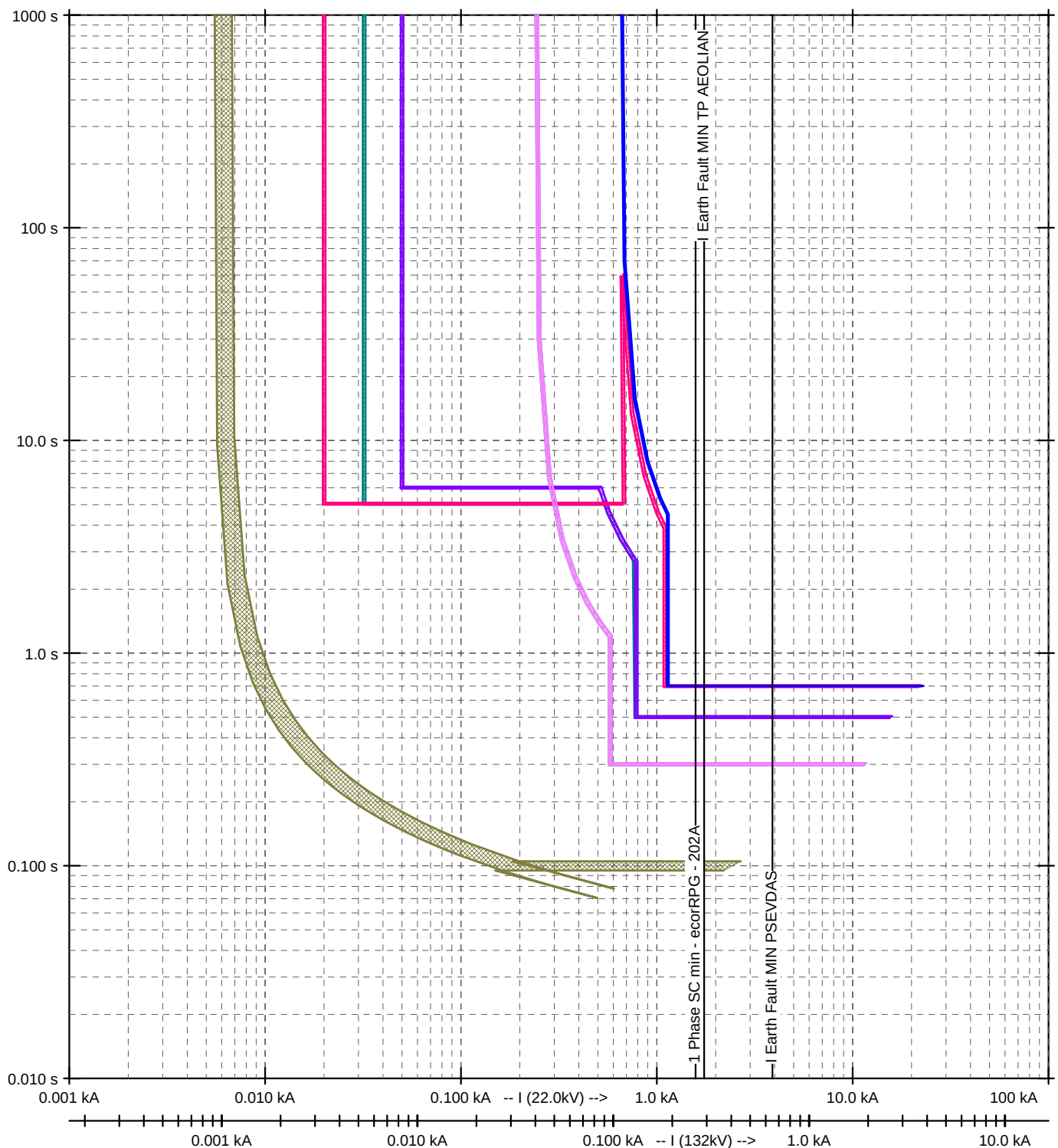
-  ekorRPG-202A - TP AEOLIAN DYNAMICS WF : CGMCOSMOS_1 - CBL WG_1 (I>)
ORMAZABAL MV Panel (VESTAS IED)
-  REF_630 - TP AEOLIAN DYNAMICS SS (-Q1) : '=AA1J00+J01 - Underground Feeder (-Q1) (I>)
Feeder from Loop 1 of TP Aeolian Dynamics WF
-  REF_630 - TP AEOLIAN DYNAMICS SS (-Q2) : '=AA1J00+J02 - Underground Feeder (-Q2) (I>)
Feeder from Loop 2 of TP Aeolian Dynamics WF
-  REF_630 - TP AEOLIAN DYNAMICS SS (-Q3) : '=AA1J00+J03 - Incomer TP Aeolian (-Q3) (I>)
Incomer from 3x(1x300mm^2) underground cable of OHL TP Aeolian
-  REF_630 - PSEVDAS SS_IPPM : '=J04+J04 - TP Aeolian Feeder (I>)
ZX1.2 MV Feeder Panel to which an OHL of Al 70mm^2 connects
-  REF_630 - PSEVDAS SS_TM : '=J03+J03 - Incomer No1 (I>)
Incomer from MV E06 Bay of Psevdas SS protecting Trafo Cables
-  REC_670 - PSEVDAS SS_HV PP : '=E06+CRP06 - Transformer Feeder (I>)
HV Control & Protection Panel Bay E06



-  ekorRPG-202A - TP AEOLIAN DYNAMICS WF : CGMCOSMOS_1 - CBL WG_1 (I>)
ORMAZABAL MV Panel (VESTAS IED)
-  REF_630 - TP AEOLIAN DYNAMICS SS (-Q1) : '=AA1J00+J01 - Underground Feeder (-Q1) (I>)
Feeder from Loop 1 of TP Aeolian Dynamics WF
-  REF_630 - TP AEOLIAN DYNAMICS SS (-Q2) : '=AA1J00+J02 - Underground Feeder (-Q2) (I>)
Feeder from Loop 2 of TP Aeolian Dynamics WF
-  REF_630 - TP AEOLIAN DYNAMICS SS (-Q3) : '=AA1J00+J03 - Incomer TP Aeolian (-Q3) (I>)
Incomer from 3x(1x300mm^2) underground cable of OHL TP Aeolian
-  REF_630 - PSEVDAS SS_IPPM : '=J04+J04 - TP Aeolian Feeder (I>)
ZX1.2 MV Feeder Panel to which an OHL of Al 70mm^2 connects
-  REF_630 - PSEVDAS SS_TM : '=J03+J03 - Incomer No1 (I>)
Incomer from MV E06 Bay of Psevdas SS protecting Trafo Cables
-  REC_670 - PSEVDAS SS_HV PP : '=E06+CRP06 - Transformer Feeder (I>)
HV Control & Protection Panel Bay E06



-  ecorRPG-202A - TP AEOLIAN DYNAMICS WF : CGMCOSMOS_1 - CBL WG_1 (I0>)
ORMAZABAL MV Panel (VESTAS IED)
-  REF_630 - TP AEOLIAN DYNAMICS SS (-Q1) : '=AA1J00+J01 - Underground Feeder (-Q1) (I0>)
Feeder from Loop 1 of TP Aeolian Dynamics WF
-  REF_630 - TP AEOLIAN DYNAMICS SS (-Q2) : '=AA1J00+J02 - Underground Feeder (-Q2) (I0>)
Feeder from Loop 2 of TP Aeolian Dynamics WF
-  REF_630 - TP AEOLIAN DYNAMICS SS (-Q3) : '=AA1J00+J03 - Incomer TP Aeolian (-Q3) (I0>)
Incomer from 3x(1x300mm^2) underground cable of OHL TP Aeolian
-  REF_630 - PSEVDAS SS_IPPM : '=J04+J04 - TP Aeolian Feeder (I0>)
ZX1.2 MV Feeder Panel to which an OHL of Al 70mm^2 connects
-  REF_630 - PSEVDAS SS_TM : '=J03+J03 - Incomer No1 (I0>)
Incomer from MV E06 Bay of Psevdas SS protecting Trafo Cables
-  REC_670 - PSEVDAS SS_HV PP : '=E06+CRP06 - Transformer Feeder (I0>)
HV Control & Protection Panel Bay E06



-  ecorRPG-202A - TP AEOLIAN DYNAMICS WF : CGMCOSMOS_1 - CBL WG_1 (I0>)
ORMAZABAL MV Panel (VESTAS IED)
-  REF_630 - TP AEOLIAN DYNAMICS SS (-Q1) : '=AA1J00+J01 - Underground Feeder (-Q1) (I0>)
Feeder from Loop 1 of TP Aeolian Dynamics WF
-  REF_630 - TP AEOLIAN DYNAMICS SS (-Q3) : '=AA1J00+J03 - Incomer TP Aeolian (-Q3) (I0>)
Incomer from 3x(1x300mm^2) underground cable of OHL TP Aeolian
-  REF_630 - PSEVDAS SS_IPPM : '=J04+J04 - TP Aeolian Feeder (I0>)
ZX1.2 MV Feeder Panel to which an OHL of Al 70mm^2 connects
-  REF_630 - PSEVDAS SS_TM : '=J03+J03 - Incomer No1 (I0>)
Incomer from MV E06 Bay of Psevdas SS protecting Trafo Cables
-  REC_670 - PSEVDAS SS_HV PP : '=E06+CRP06 - Transformer Feeder (I0>)
HV Control & Protection Panel Bay E06
-  REF_630 - TP AEOLIAN DYNAMICS SS (-Q2) : '=AA1J00+J02 - Underground Feeder (-Q2) (I0>)
Feeder from Loop 2 of TP Aeolian Dynamics WF